

IMPERIAL, NE, USA

WMO: 725626

Lat: 40.510N

Lon: 101.620W

Elev: 996

StdP: 89.92

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24091

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.7	-15.1	-21.9	0.6	-16.8	-19.2	0.8	-13.5	14.9	0.0	13.1	-0.7	3.7	250	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	37.3	19.7	35.6	19.7	33.7	19.5	22.4	30.9	21.7	30.3	21.0	29.7	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	16.8	25.1	19.2	15.7	24.5	18.4	15.0	24.3	71.0	30.7	67.8	30.1	65.2	29.7	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.3	11.6	10.3	DB	-22.6	40.2	3.4	1.7	-25.0	41.5	-26.9	42.5	-28.8	43.4	-31.3	44.7
			WB	-22.9	24.0	3.3	0.7	-25.3	24.6	-27.2	25.0	-29.0	25.4	-31.4	25.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-1.4	-0.9	5.2	9.7	15.2	21.9	25.2	23.6	18.9	10.9	4.0	-1.6		
	DBStd	10.79	6.04	6.32	5.71	5.06	4.99	3.84	2.96	3.13	4.52	5.17	5.51	5.88		
	HDD10.0	1502	353	306	165	65	13	0	0	0	2	50	189	359		
	HDD18.3	3280	611	538	407	259	121	12	1	3	47	233	431	617		
	CDD10.0	1854	1	1	17	56	175	356	472	421	269	78	8	0		
	CDD18.3	591	0	0	0	1	24	118	214	165	64	3	0	0		
	CDH23.3	7416	0	1	27	94	432	1472	2548	1813	884	140	6	0		
	CDH26.7	3686	0	0	4	22	164	708	1419	922	407	39	1	0		
(14) Wind	WSAvg	4.7	4.6	4.8	5.1	5.7	5.2	4.8	4.2	4.0	4.3	4.5	4.5	4.4		
(15) Precipitation	PrecAvg	495	12	13	30	45	76	84	73	61	37	32	17	10		
	PrecMax	755	42	55	156	171	192	222	177	218	161	107	50	32		
	PrecMin	265	0	0	2	2	2	16	13	3	0	1	0	0		
	PrecStd	101	11	12	28	32	42	49	39	43	34	27	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	21.1	26.8	29.6	34.0	38.0	40.0	38.0	36.6	30.7	23.9	17.4		
		MCWB	7.2	8.4	11.4	13.4	16.4	19.1	19.7	19.5	17.3	14.9	11.2	7.0		
	2%	DB	14.3	16.1	22.7	26.2	31.0	36.0	37.7	36.1	33.8	27.3	20.4	13.3		
		MCWB	5.8	6.5	10.0	12.8	16.6	19.2	20.0	19.9	17.9	13.7	9.4	5.3		
	5%	DB	11.0	12.5	19.4	23.0	27.9	33.2	36.1	34.0	31.5	24.0	17.1	10.3		
		MCWB	4.1	4.9	8.8	11.8	15.7	19.2	20.3	19.8	17.4	12.6	8.3	3.9		
	10%	DB	7.7	9.1	16.0	19.9	25.4	30.9	33.9	32.1	28.7	20.8	13.5	7.3		
		MCWB	2.3	3.4	7.5	10.7	14.8	18.8	20.4	19.8	16.8	11.6	6.4	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	9.1	13.4	15.4	19.6	22.9	23.7	23.0	20.8	16.6	12.2	7.9		
		MCDB	16.8	19.7	21.8	25.0	28.7	30.9	31.7	31.4	29.8	25.2	20.5	15.9		
	2%	WB	5.9	6.8	11.3	13.9	18.0	21.5	22.7	22.0	19.7	15.1	10.3	5.7		
		MCDB	13.7	15.1	20.3	22.6	26.5	29.9	31.1	30.6	28.6	23.3	18.6	12.7		
	5%	WB	4.3	5.2	9.6	12.6	16.9	20.6	21.9	21.3	18.8	13.7	8.6	4.0		
		MCDB	10.3	12.0	18.3	21.0	25.2	29.1	30.6	29.8	27.2	21.8	16.5	9.9		
	10%	WB	2.7	3.6	8.0	11.4	15.7	19.8	21.3	20.6	17.9	12.3	6.8	2.3		
		MCDB	7.4	8.9	15.6	19.1	23.5	28.7	30.1	29.0	25.8	19.4	13.3	7.1		
(35) Mean Daily Temperature Range	MDBR		13.6	13.3	15.5	14.7	14.2	15.1	15.5	15.0	16.0	15.3	14.7	13.3		
	5% DB	MCDBR	18.5	19.3	21.8	20.9	18.8	18.5	18.9	18.3	19.8	21.2	20.3	17.7		
		MCWBR	11.2	11.2	11.2	9.9	7.3	6.2	5.7	5.7	6.7	9.8	11.1	11.0		
	5% WB	MCDBR	17.5	18.4	19.9	18.2	16.3	16.2	15.8	15.5	16.3	17.8	19.2	17.2		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.264	0.290	0.317	0.344	0.357	0.379	0.369	0.335	0.296	0.266	0.256		
	taud		2.554	2.532	2.477	2.424	2.420	2.416	2.400	2.413	2.473	2.547	2.552	2.536		
	Ebn at Noon		932	969	970	957	932	915	892	893	906	916	909	901		
	Edh at Noon		70	83	98	111	115	115	116	111	98	81	69	66		
(44) All-Sky Solar Radiation	RadAvg		2.39	3.25	4.44	5.47	6.33	7.28	7.18	6.26	5.09	3.63	2.60	2.07		
	RadStd		0.17	0.24	0.36	0.30	0.43	0.36	0.27	0.29	0.23	0.33	0.16	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page